

ARCHITECTURE · EVENT PROCESSING

End-to-end flowchart covering ingestion, enrichment, decision routing, auto-remediation, and continuous learning.

START – NMS Alert Triggered



STEP 01

Event Ingestion

Source: SNMP Trap / Syslog / REST API / Webhook

Action: Validate, parse & store raw event in time-series DB

Protocols: SNMP v2c/v3 · Syslog RFC 5424 · HTTP/HTTPS · MQTT

[SolarWinds](#) [Nagios](#) [Zabbix](#) [Prometheus](#) [Datadog](#)



STEP 02

Event Enrichment

CMDB Lookup: CI details, ownership, dependencies

History: Past incidents, known issues, resolution patterns

Service Map: Affected business services & criticality

Impact: Calculate blast radius via dependency graph



STEP 03

Event Correlation & Deduplication

Dedupe: Suppress duplicates within time window

Pattern: Identify event sequences & topology paths

RCA: Correlate symptoms → root cause signal

Group: Merge related events into single incident

ML Model Correlation Engine



EVALUATE RULES • PRI • SLA • THRESHOLDS

PATH A



Ticket Creation

Auto-generate ticket with full context

Assign to appropriate team via rules

Set priority & SLA clock

ServiceNow Jira SM

PATH B



Auto Remediation

Execute diagnostic workflow (L1/L2)

Run remediation actions

Verify fix & rollback if needed

Ansible Python

PATH C



Notifications

Alert via Email / SMS

Slack & MS Teams message

PagerDuty on-call trigger

Slack PagerDuty

PATH D



Escalation

Breach SLA threshold?

Escalate to L2/L3 team

Management notification

SLA Engine



AUTO REMEDIATION PATH



STEP 05

Diagnostics – L1 & L2

L1 (Auto):

Connectivity tests · Service status check
Log analysis · Interface bounce
Cache clear · DNS flush

L2 (Deep):

Packet capture · ELAM / TCAM analysis
Memory/thread dump · I/O trace
Slow drain & fabric topology analysis

Cisco ACI

SD-WAN

SAN Switch

Servers



STEP 06

Remediation Execution

Actions: Restart services, clear cache, reset connections, bounce ports

Rollback: Automatic rollback triggered on verification failure

Safety Gate: Approval required for high-risk or production changes

Ansible Playbook

Terraform

Python Script



STEP 07

Verification & Closure

Health Check: Service metrics return to baseline

Ticket Update: Resolution summary & actions logged

CMDB Update: Config items updated post-change

Knowledge Base: Successful resolution added as article



STEP 08

Feedback & Continuous Learning

Metrics: MTTR · SLA compliance ·
Success rate

Rules: Refine correlation accuracy

Capacity: Feed forecasting models

ML Update: Train models on resolution
outcomes

